Phase D4 Final Closure: Twenty-Six Composition Pairs Establishing the Non-Obvious Multi-Commitment Governance Requirements as Prior Art, Combined Series D Scope of 176 Notes Through Phase D4, With Transition to Phase D5 Operational Test Suite Expansion

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

This note is D4.30, the thirtieth and final note in Phase D4 of Series D. It closes Phase D4 by synthesizing what the composition pair sub-series established, states the combined Series D scope through Phase D4 as 176 notes constituting the complete prior-art record for Paper 3's governance architecture, and transitions to Phase D5, which expands the operational test library from the 20-test starting point produced in D2.24 to comprehensive coverage of all governance commitments and composition pair requirements. Phase D6, which follows Phase D5, addresses boundary cases at the edges of the architecture's commitments. Together, Phases D5 and D6 add test infrastructure and edge-case analysis to a prior-art record whose substantive claims are now complete.

1. Phase D4 as the Third Layer of Series D Coverage

Series D has proceeded in three substantive layers across Phases D2, D3, and D4. Phase D2 (80 notes, #496–#575) established each of Paper 3's commitments individually — decomposing Claims 1 through 6 into operational variants and sub-commitments, each variant stated as a discrete prior-art claim. Phase D3 (30 notes, #576–#605) established what violating those commitments looks like — the failure modes that arise when an implementation omits, elides, or misreads one commitment at a time. Phase D4 (30 notes, #606–#635) establishes what specific governance scenarios require when two or more commitments must be satisfied simultaneously — the multi-commitment interactions that are non-obvious from reading either commitment in isolation.

The three layers are complementary and non-redundant. Phase D2 shows that each commitment exists and has operational content. Phase D3 shows that omitting any commitment produces a recognizable failure mode, establishing that the commitments are load-bearing rather than ornamental. Phase D4 shows that specific scenario pairings generate governance requirements not deducible from either member commitment alone, establishing that the architecture's claims extend beyond its six named claims to the interactions among them. Together the three phases provide the complete prior-art answer to three distinct classes of challenge: a claim that a commitment was already known, a claim that a commitment is unnecessary, and a claim that multi-commitment governance adds nothing beyond the individual commitments summed.

2. What Phase D4 Established: Twenty-Six Composition Pairs

Phase D4 opened with D4.01, which defined the composition pair as an analytical unit: a specific governance scenario in which two commitments interact in a way that generates requirements neither commitment specifies alone. The remaining 29 notes enumerated 26 composition pairs — 17 planned pairs (D4.02–D4.18) and 9 additional pairs identified during execution (D4.19–D4.27) — followed by D4.28 (the composition pair framework as a prior-art claim in its own right), D4.29 (an index of 25 affirmative prior-art claims extracted from the pair sub-series), and this closure note.

The 17 planned pairs (D4.02–D4.18) covered the following governance scenarios: minimum viable governance combined with time-sensitive coordination; standing configurations combined with multi-Self events; the competition variant of conflict-handling combined with the preserve tier; exit rights combined with active events; high-frequency operations combined with health monitoring; population scope combined with governance capacity; DNA absorption combined with aspect restructuring; conflict carry-through combined with vertical evolution; regulatory audit combined with documentation standards; knowledge transfer combined with DNA absorption; competitive intelligence combined with post-mortem processing; cross-organizational agreement combined with standing configuration; emergency dissolution combined with the evolution feed; partial withdrawal combined with the conflict registry; asymmetric Selves combined with standing configurations; high-frequency operations combined with cross-organizational agreement; and population participation combined with exit rights.

The 9 additional pairs (D4.19–D4.27) extended coverage to: trust calibration combined with DNA absorption; observational studies combined with governance records; home perimeter combined with contribution scope; versioning combined with process disputes; multi-domain configuration combined with cross-organizational agreement; the instinct/reasoning separation combined with exchange bounding; determinism combined with non-delegation; health indicators combined with post-mortem processing; and non-specialist governance combined with the governance floor.

The synthesis notes (D4.28–D4.29) stated the composition pair framework itself as a prior-art record and indexed the affirmative claims the 26 pairs collectively produce.

Across the 26 pairs, the recurring structural insight is the same: two governance commitments that are coherent and well-specified individually, when required simultaneously in a single governance scenario, generate constraints on sequencing, authority scope, record-keeping, or conflict handling that neither commitment states alone. The prior-art value of Phase D4 is not in naming the individual commitments — Phase D2 did that — but in showing that their interactions are determinate and specifiable rather than left to implementer discretion.

3. Combined Series D Scope Through Phase D4

The phases completed through D4.30 accumulate as follows:

- **Phase D0** (6 notes, #460–#465): Paper 3 claim-level anchor notes, one per each of the six named claims.
- **Phase D1** (30 notes, #466–#495): Foundational sub-commitments decomposed from each claim.
- **Phase D2** (80 notes, #496–#575): Operational variants and decompositions for each sub-commitment.
- **Phase D3** (30 notes, #576–#605): Anti-pattern formalizations for each commitment category.
- **Phase D4** (30 notes, #606–#635): Composition pairs and multi-commitment governance requirements.

Combined Series D through Phase D4: 176 notes.

This total represents the complete substantive prior-art record for Paper 3's governance architecture. The 176 notes cover what the architecture requires (D0–D2), what it prohibits (D3), and what specific multi-commitment scenarios demand (D4). No governance commitment named in Paper 3, no failure mode of those commitments, and no non-obvious interaction between two commitments is left without a corresponding prior-art claim in the deposited series.

The 176-note milestone is the point at which the core defensive-publication mission of Series D is complete. Every architectural claim in Paper 3 that a subsequent patent application could attempt to assert as novel has now been published as prior art in an individually citable, time-stamped, openly licensed form. Phases D5 and D6, which follow, are additive: they extend the series into test infrastructure and boundary-case analysis without opening new substantive territory.

4. Transition to Phase D5: Operational Test Suite Expansion

Phase D5 (~15 notes, #636–#650) produces the comprehensive operational test library for Paper 3. The test library serves a distinct purpose from the prior-art claims in Phases D0–D4: where those notes establish that a governance commitment exists and has operational content, the test library provides practitioners with the instruments to verify whether a given implementation satisfies the commitment.

The starting point for the Phase D5 library is D2.24, which produced a 20-test operational test suite as part of Phase D2's coverage of one governance commitment. That suite demonstrated the form — each test states a scenario, specifies an observable, and states what a compliant implementation must produce — but covered only the sub-commitments addressed in D2.24 and did not extend to composition pair requirements. Phase D5 will:

- Add tests for governance commitments addressed in D2 notes other than D2.24, filling gaps in coverage where a commitment was documented in prior-art form but not yet expressed as a testable observable.
- Add tests specifically for composition pair requirements identified in Phase D4 — scenarios where two commitments must be satisfied simultaneously and where compliance requires observable behavior that goes beyond satisfying either commitment individually.
- Add trilogy-wide tests that verify consistency conditions across Paper 3's commitments as a set, including conditions whose violation would not register as a failure of any single commitment in isolation.

The Phase D5 library is what converts the 176-note prior-art record into a practitioner-facing compliance instrument. A practitioner implementing Paper 3's governance architecture who completes the Phase D5 test battery will have verified, by observable behavior, that their implementation satisfies the architecture's requirements. A practitioner whose implementation fails one or more tests will have a precise description of which commitment is not satisfied and what observable correction would bring the implementation into compliance.

5. Phase D6 Preview: Boundary Cases

Phase D6 (~15 notes, #651–#665) addresses governance at the edges of Paper 3's commitments. Where Phases D0–D4 established the architecture in its normal operating range and Phase D3 established failure modes, Phase D6 examines what the architecture requires at three specific edges:

The minimum edge: what the architecture requires when a single aspect participates in a single event under minimal governance. These are the conditions under which the architecture's commitments are non-trivially binding even at the smallest possible scale, establishing that minimum-viable implementations cannot escape the commitments by reducing scope.

The maximum edge: what the architecture requires under full-Self contribution, many simultaneous events, and maximum governance depth. These are the conditions under which the architecture's commitments remain tractable, establishing that the architecture does not degrade into unmanageability at scale.

The explicit-choice edge: the conditions at which the architecture requires implementers to make an explicit architectural decision rather than defaulting — where two governance commitments are compatible but not jointly satisfiable by a single default policy, and where the implementer must choose which satisfaction path to take and record that choice as substrate content.

Phase D6 boundary cases do not introduce new commitments. They establish that the commitments already stated in Phases D0–D4 apply at the architecture's edges in determinate and verifiable ways, completing the boundary analysis that accompanies any well-specified governance architecture.

6. Series D Completion Trajectory

With Phase D4 complete and 176 notes deposited, Series D's completion trajectory through its remaining phases is:

- Phase D5: ~15 notes (#636–#650), operational test library
- Phase D6: ~15 notes (#651–#665), boundary cases
- **Series D projected total: ~206 notes**

Beyond Series D, the derivation series continues with Series CC (Paper 3 cross-derivation against Papers 1 and 2, ~40 notes) and Series T (trilogy-wide concept disambiguation, ~50 notes). The combined defensive publication record across all series — Series A for Paper 1, Series B for Paper 2, Series C for the Paper 1/Paper 2 cross-derivation, Series D for Paper 3, Series CC, and Series T — converges on a comprehensive prior-art record for the full three-paper CKS theory architecture.

7. Summary

D4.30 closes Phase D4 with the following record: 26 composition pairs establishing the non-obvious multi-commitment governance requirements of Paper 3's architecture as prior art, built across 30 notes spanning D4.01 through D4.29 plus this closure note. Combined with Phases D0 through D3, Series D has now deposited 176 notes constituting the complete substantive prior-art record for Paper 3. Phase D5 will expand the operational test library from D2.24's 20-test starting point to comprehensive coverage of all governance commitments and composition pair requirements. Phase D6 will complete the boundary-case analysis. The core defensive-publication mission of Series D — establishing that no commitment, failure mode, or multi-commitment interaction in Paper 3's governance architecture is novel — is complete at D4.30.

End of D4.30. Phase D4 closed. Series D at 176 notes. Transition to Phase D5.